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Inventor(s)	Palaskar; Ram et al.

Lock detection mechanism for electronic locking differential

Abstract

The present disclosure relates to an electronic locking differential gear mechanism having a lock detection mechanism for sensing and indication of a locked and unlocked state of the locking mechanism. The electronic locking differential gear mechanism includes an electronic stator which, when activated, rotates a ramped collar about an axis of the mechanism. The ramps of the ramped collar axially displace a plurality of push rods, which displace a locking collar and cause it to engage with a locker gear, placing the mechanism in a locked state. A sensor assembly detects the displacement of a sensing plate which is displaced along with the push rods, and thereby senses whether the mechanism is in a locked or unlocked state.

Inventors:	**Palaskar; Ram (Pune, IN), Dube; Piyush (Nagpur, IN), Edler; Andrew (Homer, MI)**
Applicant:	**Eaton Intelligent Power Limited** (Dublin, IE)
Family ID:	**84331832**
Assignee:	**Eaton Intelligent Power Limited (Dublin, IE)**
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Primary Examiner: Wu; Lori

Attorney, Agent or Firm: Merchant & Gould P.C.

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application is a National Stage Application of PCT/EP2022/025434, filed on Sep. 15, 2022, which claims priority to India application No. 202111041577, titled LOCK DETECTION MECHANISM FOR ELECTRONIC LOCKING DIFFERENTIAL, filed on Sep. 15, 2021, the disclosures of which are incorporated herein by reference in their entireties. To the extent appropriate, a claim of priority is made to each of the above disclosed applications.

BACKGROUND OF THE INVENTION

(2) Vehicle drivetrains may include differential mechanisms which include gear sets that allow torque to be applied differently to wheels of the same axle during operation of the vehicle. In some instances, it may be desired to include a locking mechanism in the differential so that all torque is delivered equally to both wheels of the axle. Locking differentials, including electronic locking differentials, are generally known in the art. For locking differentials, it may be desirable to have a detection structure on such mechanisms to determine the locking state of the differential and provide the locking state to the operator of a vehicle.

SUMMARY

(3) In some aspects, a locking mechanism is included in the differential mechanism so that all available torque is delivered equally to both wheels of the axle, locking both wheels in synchronous rotation with each other. A detection structure on such a locking mechanism to determine the locking state of the differential and provide the locking state to the operator of a vehicle may provide useful benefits to the driver, including ensuring that the locking mechanism is functioning properly, and monitoring vehicle performance and status for safe vehicle operations.

(4) Accordingly, the present application describes, in an example aspect, a differential assembly comprising a first housing; a first half-axle output; a second half-axle output; a power input location; a gear set; an electrically activated locking assembly; and a sensor assembly. In an aspect, the gear set is disposed within the first housing and operably connects the first half-axle output, the second half-axle output, and the power input location together. In an aspect, the electrically activated locking assembly includes a second housing mounted to the first housing and is operable

between a locked position, in which the first and second axle outputs are prevented from rotating relative to each other, and an unlocked position, in which the first and second half-axle outputs are enabled to rotate relative to each other. In an aspect, the sensor assembly includes a position sensing plate and a position sensor, the position sensing plate being coupled to the locking assembly such that operation of the locking assembly between the locked and unlocked positions causes axial movement of the position sensing plate and such that the position sensing plate is non-rotatable with respect to the second housing, and the position sensor is configured to measure an axial position of the position sensing plate.
(5) In an aspect, the locking assembly includes a stator mounted within the second housing, a rotatable ramped collar actuated by the stator, a plurality of push rods axially displaceable by rotation of the ramped collar, and a locking collar axially displaceable by movement of the plurality of push rods, where in the locked position, the locking collar is engaged with a locker gear of the gear set, and where in the unlocked position, the locking collar is disengaged with the locker gear.
(6) In an aspect, the ramped collar is located axially between the stator and the position sensing plate.
(7) In an aspect, the position sensing plate is secured to the plurality of push rods.
(8) In an aspect, one or more of the plurality of push rods includes a groove for receiving an edge portion of the position sensing plate. In an aspect, the edge portion defines a central aperture of the position sensing plate.
(9) In an aspect, the position sensing plate is secured to the plurality of push rods with one or more fasteners. In an aspect, the one or more fasteners are snap rings.
(10) In an aspect, the position sensing plate includes an axially extending tab and the position sensor is configured to sense movement of the axially extending tab.
(11) In an aspect, the sensor assembly includes a sensor target mounted to the axially extending tab.
(12) In an aspect, the position sensor is a magnetic-type position sensor that is spaced from the position sensing plate.
(13) In an example aspect, a lock detection mechanism for a differential assembly comprises a locking element of the differential assembly; one or more push rods displaceable along an axis of the differential assembly, connected at a first end to a locking collar of the locking element and in contact at a second end with a ramped collar; a position sensing plate axially displaceable along the axis, wherein the position sensing plate is displaceable by the one or more push rods; and a sensor assembly to detect a status of the locking element, wherein the sensor assembly detects a displacement of the position sensing plate. In an aspect, the locking element is operable between a locked state and an unlocked state. In an aspect, the ramped collar is rotatable about the axis. In an aspect, the ramped collar comprises one or more ramp valleys and one or more ramp peaks. In an aspect, the one or more push rods are displaceable by the one or more ramp valleys and one or more ramp peaks.
(14) In an aspect, the lock detection mechanism further comprises a sensor of the sensor assembly; and a tab of the sensor assembly. In an aspect, the tab is connected to the position sensing plate at a first end. In an aspect, the tab is displaceable by the position sensing plate along the axis. In an aspect, the tab is connected to a sensor target at a second end. In an aspect, a position of the sensor target is detectable by the sensor.
(15) In an aspect, the sensor is a magnetic-type position sensor that is spaced from the position sensing plate.
(16) In an aspect, the position sensing plate comprises a notch, and at least a portion of the tab fits within the notch.
(17) In an aspect, the position sensing plate comprises one or more recessed portions, and each of the one or more recessed portions receives a portion of one of the one or more push rods.
(18) In an aspect, each of the one or more push rods includes a groove receiving the respective recessed portion of the position sensing plate and at least one snap-ring on at least one side of the

position sensing plate.
(19) In an aspect, the ramped collar is rotatable by an electronic actuator mechanism.
(20) An example method of detecting a locked state of a differential assembly, comprises rotating, about an axis of a differential assembly, a ramped collar; displacing a push rod along the axis (the push rod in contact with the ramped collar at a first end) by the rotation of the ramped collar; displacing a locking collar of a locking element along the axis by the displacement of the push rod, where the locking collar is connected to the push rod at a second end; displacing a position sensing plate along the axis, by the displacement of the push rod; and sensing, by a sensor assembly, an axial displacement of the position sensing plate.
(21) In an aspect, the method further comprises displacing a tab along the axis, by the displacement of the position sensing plate, where the tab is connected at a first end to the position sensing plate; detecting a position of a second end of the tab by a sensor of the sensor assembly; and indicating a status of the locking element.
(22) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Non-limiting and non-exhaustive examples are described with reference to the following Figures.
(2) FIG. **1** illustrates a perspective view of an electronic locking differential gear mechanism, constructed in accordance with principles of this disclosure, according to an example.
(3) FIG. **2** illustrates a perspective view of the electronic locking differential gear mechanism of FIG. **1**, with a housing removed, according to an example.
(4) FIG. **3** illustrates a left-side view of the electronic locking differential gear mechanism of FIG. **1**, according to an example.
(5) FIG. **4** illustrates a cross-sectional view of the electronic locking differential gear mechanism of FIG. **2**, along the plane indicated in FIG. **3**, according to an example.
(6) FIG. **5** illustrates a front-perspective exploded view of the electronic locking differential gear mechanism of FIG. **2**, according to an example.
(7) FIG. **6** illustrates a rear-perspective exploded view of the electronic locking differential gear mechanism of FIG. **2**, according to an example.
(8) FIG. **7** illustrates a perspective view of a sensor assembly for an electronic locking differential gear mechanism, constructed in accordance with principles of this disclosure, according to an example.
(9) FIG. **8** illustrates an exploded view of the sensor assembly of FIG. **7**, according to an example.
(10) FIG. **9** illustrates a side view of a sensing plate for the sensor assembly of FIG. **7**, according to an example.
(11) FIG. **10** illustrates a front view of the electronic locking differential gear mechanism of FIG. **2**, in an unlocked state, according to an example.
(12) FIG. **10**A illustrates a front view of an alternate embodiment of the electronic locking differential gear mechanism of FIG. **10**, in an unlocked state, with two snap rings engaged with each push rod, according to an example.
(13) FIG. **11** illustrates a front view of the electronic locking differential gear mechanism of FIG. **2**, in a locked state, according to an example.
(14) FIG. **12** illustrates a schematic view of a vehicle within which the electronic locking

differential gear mechanism may be used.
DETAILED DESCRIPTION
(15) In the following detailed description, references are made to the accompanying drawings that form a part hereof, and in which are shown, by way of illustrations, specific embodiments or examples. These aspects may be combined, other aspects may be utilized, and structural changes may be made without departing from the present disclosure. Examples may be practiced as methods, systems, or devices. The following detailed description is therefore not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims and their equivalents.
(16) Vehicle drivetrains may include differential mechanisms which include gear sets **50** that allow torque to be applied differently to wheels of the same axle during operation of the vehicle. In some aspects, a driveshaft transferring torque from an engine or motor may, via a drive pinion gear, transfer torque to a ring gear of a transmission (for example, mounted to a flange of the transmission) by engagement with the ring gear. This ring gear is in a meshing engagement with a plurality of inner pinion gears (or spider gears) **52**. In an aspect, the inner pinion gears **52** are in a meshing engagement with two side gears **54**, **56**, each side gear additionally having internal splines **58**, **60** corresponding to a half-axle output. In an aspect, the ring gear is located outside of a case or housing, and the gear set (which operably connects the half-axle outputs and the power input location (driveshaft) together), including the inner pinion gears **52** and side gears **54**, **56**, are located within the case/housing.
(17) In some aspects, a locking mechanism is included in the differential mechanism so that all available torque is delivered equally to both wheels of the axle, locking both wheels in synchronous rotation with each other. In some aspects, the locking mechanism is moved between a locked state and an unlocked state by an electrical action. In some aspects, the electrical action includes an electric stator. In some aspects, a driver or other user of a vehicle may trigger the electrical locking mechanism by way of a button, switch, or similar interactive device.
(18) Referring to FIG. **12**, an example vehicle driveline is shown in accordance with the above description and disclosure and within which the disclosed electronic locking differential gear assembly **10** may be used. As shown, the vehicle driveline is provided with a power plant **106**, such as an internal combustion engine or electric motor, a transmission **107**, a power transfer unit **108** that are operatively connected at the front of the vehicle to transmit torque directly to a front left axle **100**, and front right axle **101** such that wheels **102** and **103** receive torque to provide traction to the vehicle via wheel hubs **115** and **116**. Via mechanisms in the power transfer unit **108**, such as a hypoid gear and pinion, the drive shaft **109** receives torque and transmits it to the rear of the vehicle. As shown, an all-wheel drive coupling **120** connects to the drive shaft **109**, and a rear drive unit **110** may house the disclosed locking differential assembly **10** within a rear differential assembly. As discussed further herein, the locking differential assembly **10** may be operated in either an open or locked mode. In the open mode, the left rear wheel **113**, via wheel hub **117** and left rear axle **111**, can spin at a speed that is different from the right wheel **114**. Likewise, the right rear wheel **114**, via wheel hub **118** and right rear axle **112**, can spin at a different speed than left rear wheel **113**. In the locked mode, both left and right rear wheels **113** and **114** receive the same torque because the left and right rear axle **111** and **112** are locked together to prevent relative rotation via internal components in the rear differential. An example of a differential mechanism is described in further detail in U.S. Pat. No. 9,657,827 B2, the entire disclosure of which, except for any definitions, disclaimers, disavowals, and inconsistencies, is incorporated herein by reference.
(19) Referring now to the Figures, there is shown an electronic locking differential gear mechanism (e.g., differential assembly) **10** that includes cam/ramp actuated direct acting electronic locking capability. The gear set **50** of the electronic locking differential gear mechanism **10** is located within a case **20** (refer to FIG. **1**), and in some examples, torque may be transferred to the gear set **50** via a ring gear mounted to a flange **23**. Accordingly, the flange **23** may be referred to as a power

input location.
(20) In an aspect, an electric actuator mechanism (e.g., electrically actuated locking assembly or locking element) **12** of the electronic locking differential gear mechanism **10** includes a stator housing **40**. The stator housing **40** is mounted to case (e.g., housing) **20**. The stator housing **40** is operable between a locked position, in which first and second half-axle outputs are prevented from rotating relative to each other, and an unlocked position, in which the first and second half-axle outputs are enabled to rotate relative to each other. Stator housing **40** includes an electric stator **16**, which, in some aspects, causes a ramped collar **18** to be rotationally actuated in relation to the stator housing **40**. In an aspect, ramped collar **18** rotates about an axis **2**.
(21) Referring to FIG. **10-11**, in an aspect, ramped collar **18** includes one or more ramps (for example, one or more locations axially where it has a greater or lesser length). Ramped collar **18** includes one or more ramp valleys **17**, where the axial length of the ramped collar **18** is the least. Ramped collar **18** includes one or more ramp peaks **19**, where the axial length of the ramped collar **18** is the greatest.
(22) Electric actuator mechanism **12** includes at least one actuating push rod **14**. Although three push rods **14** are depicted in the example embodiments of the Figures, other embodiments may include more than three push rods **14** or fewer than three push rods **14**. Push rods **14** are operably coupled to electric stator **16** via the ramped collar **18**. In some aspects, push rods **14** may be generally cylindrical or rod-like in shape. Push rods **14** contact ramped collar **18** at an end of the push rods **14**. In some aspects, push rods **14** pass through the case **20** in a bore, and do not rotate with relation to the case **20**.
(23) Push rods **14** are coupled to a locking collar **24**. In some examples, push rods **14** are coupled to locking collar **24** via press fit design, threaded connection, or any suitable fastening or attachment means.
(24) Push rods **14** are axially displaceable along (parallel to) the axis X of the differential assembly **10**. Push rods **14** are axially displaceable by the rotation of the ramped collar **18**. Locking collar **24** is axially displaceable by the displaced movement of the push rods **14**. By use of the term axial or axially displaceable, it is meant to define a direction that is parallel to the longitudinal axis X of the differential assembly **10**.
(25) Locking collar **24** includes teeth **26** formed on an inner diameter of the locking collar **24**, which selectively engage with external teeth **27** of a locker gear **28**. When the teeth **26** of locking collar **24** engage with the external teeth **27** of locker gear **28**, the electronic locking differential gear mechanism **10** is placed into a locked position. When the teeth **26** of locking collar **24** are disengaged from the external teeth **27** of locker gear **28**, the electronic locking differential gear mechanism **10** is placed into an unlocked position.
(26) The locking collar **24** is disposed within a cavity defined at an outer boundary by the gear case **20** and defined at an inner boundary by the locker gear **28**.
(27) In some aspects, the locking collar **24** is biased to a disengaged or unlocked position by a spring or other biasing force member.
(28) In an aspect, each of the push rods **14**, in contact with ramped collar **18** at an end, is at an initial position (corresponding to an unlocked state) with its end in contact with a ramp valley **17**. As ramp collar **18** is rotated about axis **2**, each of the push rods **14** is displaced axially toward locker gear **28**, as the end in contact with ramped collar **18** is moved to a maximally displaced position in contact with a ramp peak **19** (corresponding to a locked state). This displaces locking collar **24** toward locker gear **28**, allowing teeth **26** and external teeth **27** to engage.
(29) The electronic locking differential gear mechanism **10** includes a sensor assembly **32** to determine the locking state/status (for example, locked or unlocked) of the electronic locking differential gear mechanism **10**. The sensor assembly **32** includes at least a sensing plate **34** (e.g., position sensing plate). Sensing plate **34** is axially displaceable along the axis **2**. Sensing plate **34** is axially displaceable by at least one of the push rods **14**.

(30) Sensing plate **34** is attached/secured to at least one of the push rods **14**. In an aspect, one or more of the push rods **14** includes a groove **15** for receiving an inner edge portion **37** of the sensing plate **34**. In an aspect, the sensing plate **34** is attached to a push rod **14** using one or more fasteners. In an aspect, the sensing plate **34** is attached to a push rod **14** using one or more snap rings **36**. In an aspect, the sensing plate **34** is attached to a push rod **14** using a step (for example, of a groove **15**) on one side of the push rod **14**. In an aspect, the sensing plate **34** is attached to a push rod **14** using a step of groove **15** and a snap ring **36** (on a side of the sensing plate **34** opposite the step). In an aspect, the sensing plate **34** is attached to a push rod **14** by sitting within groove **15** and having a snap ring **36** on either side of the sensing plate **34** (refer to FIG. **10**A). In an aspect, groove **15** and/or one or more snap rings **36** retain the sensing plate **34** relative to the push rod **14**. Other types of fasteners may also be used.
(31) In an aspect, sensing plate **34** includes one or more recesses **33**. Recesses **33** are located on the inner edge portion **37** of sensing plate **34**. In some aspects, the number of recesses **33** equals the number of push rods **14**. In some aspects, recesses **33** are semi-circular (or another segment of a circle) in nature. In some aspects, each of the recesses **33** engage with (for example, receive) at least a portion of an outer surface of one of the push rods **14**. In some aspects, each of recesses **33** engage with one of the push rods **14** at a groove **15**.
(32) In an aspect, the inner edge portion **37** of sensing plate **34** defines a central aperture **39** of the sensing plate **34**.
(33) Sensing plate **34** is thereby coupled to the electric actuator mechanism (e.g., locking assembly) **12** such that operation of the locking assembly **12** between the locked and unlocked positions causes axial movement of the sensing plate **34**.
(34) Sensing plate **34** is coupled to the electric actuator mechanism (e.g., locking assembly) **12** such that sensing plate **34** is non-rotatable with respect to the stator housing.
(35) In an aspect, the ramped collar **18** is located axially between the electric stator **16** and the sensing plate **34**.
(36) The sensor assembly **32** includes sensor electronics **38**. Sensor electronics **38** include a position sensor configured to measure an axial position of the sensing plate **34** (e.g., detects a displacement of the sensing plate **34**). In an example, sensor assembly **32** is attached or mounted to the stator housing **40** by any appropriate means of attachment. Various sensor types may be utilized including, magnetic-type sensors such as Hall-effect sensors. In an aspect, the sensor is spaced from the sensor tab **42**, discussed below, of the sensing plate **34** such that the sensor does not physically contact the axially displaceable tab **42**. Accordingly, the sensor assembly **32** may be referred to as a non-contacting sensor assembly. In an embodiment, a linear variable differential transformer (LVDT) sensor may be utilized.
(37) In an aspect, the sensor assembly **32** includes a sensor tab **42**. Sensor tab **42** is displaceable by the sensing plate **34** along axis **2**. Sensor tab **42** extends axially from the sensing plate **34** in a direction away from the locker gear **28**. In an aspect, sensor tab **42** has a substantially L-shaped cross section, with the shorter base length of the L-shape contacting a surface of the sensing plate **34** opposite of the sensor electronics. In an aspect, sensor tab **42** is attached to the sensing plate **34**. Sensor tab **42** may be attached to, mounted on, or fastened to sensing plate **34** by any means, including by welding, the use of adhesives, or the use of fasteners. In another aspect, sensor tab **42** and sensing plate **34** may be integrally formed as a single component. In an aspect, sensing plate **34** includes a notch **35**. In some embodiments, at least a portion of tab **42** may engage with (for example, sit in or fit in) notch **35**.
(38) In an aspect, sensor tab **42** includes a sensor target **44** attached to or mounted thereon. The sensor tab **42** is positioned adjacent to the sensor electronics **38** to determine axial movement of the sensor plate **34**. In an aspect, a position of the sensor target **44** is detectable by the sensor assembly. In an aspect, the sensor target **44** is a magnet and the sensor electronics **38** are configured to detect a magnetic field of the sensor target **44**.

(39) In an aspect, stator housing **40** includes anti-rotation lugs or tabs **46**, **47** formed or attached thereon. The sensor tab **42** is positioned between the anti-rotation lugs tabs **46**, thereby preventing rotation of the sensing plate **34** relative to the stator housing **40**. To further provide anti-rotation capability, an additional extension or tab structure **43** is shown as being part of or secured to the sensing plate **34** and located oppositely from the sensor tab **42**. The tab structure **43** interacts with lugs **47** to prevent rotation of the sensing plate **34**. By providing two such structures, the anti-rotation forces at tabs **42**, **43** are balanced such that an undesirable torque exerted on the sensing plate **34** by the lugs is reduced or eliminated. As with sensor tab **42** and sensing plate **34**, the tab structure **45** may also be attached to, mounted on, or fastened to sensing plate **34** by any means, including by welding, the use of adhesives, or the use of fasteners, or may be integrally formed as a single component with the sensing plate.
(40) In an example use, the electric actuator mechanism **12** is in a normal, unlocked status. Refer to FIG. **10**. For example, it may be biased to an unlocked state by a spring or biasing element. In the unlocked state, locking collar **24** is moved axially away from engagement with locker gear **28**.
(41) Upon activation/energizing (which may be in response to a control signal from a control module or may be actuated in response to activation of a switch by a user) of the electric stator **16**, the ramped collar **18** is rotated about axis **2**, and at least one push rod **14** is displaced axially. As shown at FIGS. **5** and **6**, a bearing assembly **17** can be provided between the stator **16** and collar **18** to facilitate relative movement between the two components. Locking collar **24** is moved by the displacement of the one or more push rods **14** to engage with the locker gear **28**, which causes the locking of the electronic locking differential gear mechanism **10**. Refer to FIG. **11**.
(42) Sensing plate **34** is displaced axially with the push rods **14** and correspondingly the sensor tab **42** with the sensor target **44** is displaced relative to the sensor electronics **38**. The position sensor senses/detects the axial displacement of the sensor target **44** (and therefore the displacement of the sensing plate **34**).
(43) A signal is thereby produced that indicates that the electronic locking differential gear mechanism **10** is in a locked condition. The status can then be indicated to a user of the vehicle, in some aspects.
(44) In some aspects, after the electric stator **16** is de-energized, the electronic locking differential gear mechanism **10** is returned to an unlocked position (for example, by a biasing element or spring). The sensing plate **34** moves with the push rods **14** (and correspondingly the sensor tab **42** with the sensor target **44** moves relative to the sensor electronics **38**). A signal is thereby produced that indicates that the differential **10** is in an un-locked condition. The status can then be indicated to a user of the vehicle, in some aspects.
(45) For the purposes of this application, terms such as “left,” “right,” “front,” “back,” “upper,” “lower,” “upward,” and “downward” are intended to be descriptive with reference to and in relation to the orientation shown in the Figures for clarity and are not meant to be limiting, but the examples as practiced and included in the scope of the claims may include examples where the systems and devices are in a different orientation.
(46) While particular uses of the technology have been illustrated and discussed above, the disclosed technology can be used with a variety of environments in accordance with many examples of the technology. The above discussion is not meant to suggest that the disclosed technology is only suitable for implementation within the environments shown and described above. For examples, while certain technologies described herein were primarily described in the context of sealed battery cases, technologies disclosed herein are applicable to sealed components generally.
(47) This disclosure described some aspects of the present technology with reference to the accompanying drawings, in which only some of the possible aspects were shown. Other aspects can, however, be embodied in many different forms and should not be construed as limited to the aspects set forth herein. Rather, these aspects were provided so that this disclosure was thorough

and complete and fully conveyed the scope of the possible aspects to those skilled in the art.
(48) As should be appreciated, the various aspects described with respect to the Figures herein are not intended to limit the technology to the particular aspects described. Accordingly, additional configurations can be used to practice the technology herein and/or some aspects described can be excluded without departing from the methods and systems disclosed herein.
(49) Similarly, where operations of a process are disclosed, those operations are described for purposes of illustrating the present technology and are not intended to limit the disclosure to a particular sequence of operations. For example, the operations can be performed in differing order, two or more operations can be performed concurrently, additional operations can be performed, and disclosed operations can be excluded without departing from the present disclosure. Further, each operation can be accomplished via one or more sub-operations. The disclosed processes can be repeated.
(50) Although specific aspects were described herein, the scope of the technology is not limited to those specific aspects. One skilled in the art will recognize other aspects or improvements that are within the scope of the present technology. Therefore, the specific structure, acts, or operations are disclosed only as illustrative aspects. The scope of the technology is defined by the following claims and any equivalents therein.

Claims

1. A differential assembly comprising: a) a first housing; b) a first half-axle output; c) a second half-axle output; d) a power input location; e) a gear set disposed within the first housing and operably connecting the first half-axle output, the second half-axle output, and the power input location together; f) an electrically activated locking assembly including a second housing mounted to the first housing and being operable between a locked position, in which the first and second axle half-axle outputs are prevented from rotating relative to each other, and an unlocked position, in which the first and second half-axle outputs are enabled to rotate relative to each other, wherein the locking assembly includes a stator mounted within the second housing, a rotatable ramped collar actuated by the stator, a plurality of push rods axially displaceable by rotation of the ramped collar, and a locking collar axially displaceable by movement of the plurality of push rods, wherein in the locked position, the locking collar is engaged with a locker gear of the gear set, wherein in the unlocked position, the locking collar is disengaged with the locker gear; g) a sensor assembly including a position sensing plate and a position sensor, the position sensing plate being coupled to the locking assembly such that operation of the locking assembly between the locked and unlocked positions causes axial movement of the position sensing plate and such that the position sensing plate is non-rotatable with respect to the second housing, the position sensor being configured to measure an axial position of the position sensing plate.
2. The differential assembly of claim 1, wherein the ramped collar is located axially between the stator and the position sensing plate.
3. The differential assembly of claim 1, wherein the position sensing plate is secured to the plurality of push rods.
4. The differential assembly of claim 3, wherein one or more of the plurality of push rods includes a groove for receiving an edge portion of the position sensing plate.
5. The differential assembly of claim 4, wherein the edge portion defines a central aperture of the position sensing plate.
6. The differential assembly of claim 3, wherein the position sensing plate is secured to the plurality of push rods with one or more fasteners.
7. The differential assembly of claim 6, wherein the one or more fasteners are snap rings.
8. The differential assembly of claim 1, wherein the position sensing plate includes an axially extending tab and wherein the position sensor is configured to sense movement of the axially

extending tab.
9. The differential assembly of claim 8, wherein the sensor assembly includes a sensor target mounted to the axially extending tab.
10. The differential assembly of claim 1, wherein the position sensor is a magnetic-type position sensor that is spaced from the position sensing plate.
11. A lock detection mechanism for a differential assembly, the lock detection mechanism comprising: a locking element of the differential assembly, operable between a locked state and an unlocked state; one or more push rods displaceable along an axis of the differential assembly, connected at a first end to a locking collar of the locking element and in contact at a second end with a ramped collar, wherein: the ramped collar is rotatable about the axis, the ramped collar comprises one or more ramp valleys and one or more ramp peaks, and the one or more push rods are displaceable by the one or more ramp valleys and one or more ramp peaks; a position sensing plate axially displaceable along the axis, wherein the position sensing plate is displaceable by the one or more push rods; and a sensor assembly to detect a status of the locking element, wherein the sensor assembly detects a displacement of the position sensing plate.
12. The lock detection mechanism of claim 11, further comprising: a sensor of the sensor assembly; and a tab of the sensor assembly, wherein the tab is connected to the position sensing plate at a first end, wherein the tab is displaceable by the position sensing plate along the axis, wherein the tab is connected to a sensor target at a second end, and wherein a position of the sensor target is detectable by the sensor.
13. The lock detection mechanism of claim 12, wherein the sensor is a magnetic-type position sensor that is spaced from the position sensing plate.
14. The lock detection mechanism of claim 12, wherein the position sensing plate comprises a notch, and wherein at least a portion of the tab fits within the notch.
15. The lock detection mechanism of claim 11, wherein the position sensing plate comprises one or more recessed portions, each of the one or more recessed portions receiving a portion of one of the one or more push rods.
16. The lock detection mechanism of claim 15, wherein each of the one or more push rods includes a groove receiving the respective recessed portion of the position sensing plate and at least one snap-ring on at least one side of the position sensing plate.
17. The lock detection mechanism of claim 11, wherein the ramped collar is rotatable by an electronic actuator mechanism.
18. A method of detecting a locked state of a differential assembly, the method comprising: rotating, about an axis of a differential assembly, a ramped collar; displacing a push rod along the axis, the push rod in contact with the ramped collar at a first end, by the rotation of the ramped collar; displacing a locking collar of a locking element along the axis by the displacement of the push rod, wherein the locking collar is connected to the push rod at a second end; displacing a position sensing plate along the axis, by the displacement of the push rod; and sensing, by a sensor assembly, an axial displacement of the position sensing plate.
19. The method of claim 18, further comprising: displacing a tab along the axis, by the displacement of the position sensing plate, wherein the tab is connected at a first end to the position sensing plate; detecting a position of a second end of the tab by a sensor of the sensor assembly; and indicating a status of the locking element.